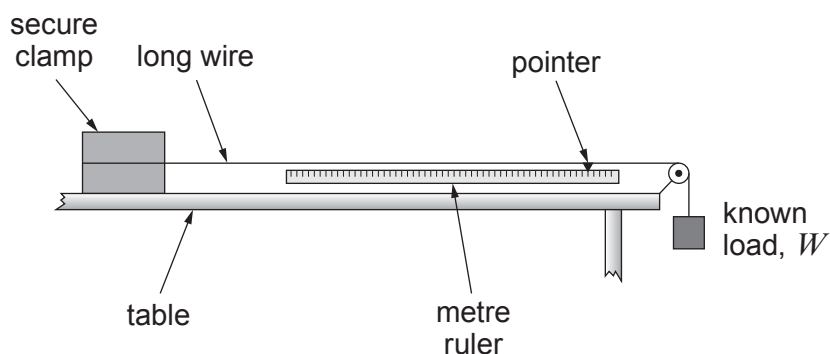


4. Karen is an A level Physics student who has been absent for some of her Physics practical lessons. During her absence, the other students in the class have been using the apparatus below to identify the metal in a wire by finding the Young modulus of the metal and comparing it with known values.



In order to catch up with the other students, Karen decides to use the same apparatus to carry out a quick experiment to find the Young modulus. Her notes are shown in the box below.

- Take single reading of diameter using a micrometer of resolution 0.01 mm.  
**Diameter = 0.18 mm**
- Use a metre ruler of resolution 1 mm to measure the initial length of the wire.  
**Initial length = 1.940 m**
- Set up the apparatus and add a **mass of 1.0 kg** to the end of the wire.  
Measure extension.  
**Extension =  $6.0 \pm 0.5$  mm**

(a) Use Karen's notes to answer the following questions.

- (i) Calculate the cross-sectional area of the wire in  $\text{m}^2$  and show that the **percentage** uncertainty in its value is approximately 11%. [2]

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- (ii) Show, with an appropriate calculation, that the uncertainty in the length reading can be considered negligible. [1]

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- (b) (i) Determine the Young modulus of the metal. [2]

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- (ii) Karen is told by her teacher that the percentage uncertainty in the load is 5%. Use this information along with Karen's notes and your answers to (a) to calculate the **absolute** uncertainty in the Young modulus. Give your answer in GPa and to an appropriate number of significant figures. [4]

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- (c) The following table shows Young modulus values of the wires available to the students.

| Metal     | Young modulus / GPa |
|-----------|---------------------|
| aluminium | 70                  |
| zinc      | 100                 |
| bronze    | 103                 |
| copper    | 117                 |
| nickel    | 166                 |

Karen's friend Jack determined the Young modulus of the metal to be  $126 \pm 20$  GPa. They both use the above table to determine the metal in their wires. Discuss how Jack's conclusions differ from Karen's. [2]

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- (d) Jack had more time to carry out this experiment than Karen. Suggest differences in the way Jack carried out the experiment (i.e. procedure) **and** in the way that he analysed his results which led to him reaching a different conclusion to Karen. [3]

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Examiner  
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